Malkin Textbook Responses – Final Report

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1 2-4 Malkin

Explain why the value θ_K, entering the Kohlrausch function, Eq. 2.1.6 is not a relaxation time. How do you find relaxation times for this relaxation function?

$$\varphi(t) = \exp\left[-\left(\frac{t}{\theta_K}\right)^n\right], \qquad \varphi(t) = \int_0^\infty G(\theta)\, e^{-t/\theta}\, d\theta.$$

where $G(\theta)$ is the relaxation-time spectrum (what we want to find). Let $s = 1/\theta$, so $d\theta = -ds/s^2$,

$$\varphi(t) = \int_0^\infty \frac{G(1/s)}{s^2}\, e^{-st}\, ds \equiv \int_0^\infty \tilde{G}(s)\, e^{-st}\, ds, \qquad \tilde{G}(s) = \frac{G(1/s)}{s^2}.$$

Therefore $\tilde{G}$ is obtained by an inverse Laplace transform:

$$\tilde{G}(s) = \mathcal{L}^{-1}\{\varphi(t)\}(s) = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty} e^{st}\, \varphi(t)\, dt$$

$$\boxed{\tilde{G}(s) = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty} e^{st}\, e^{-(s_K t)^n}\, dt, \quad \text{where } s_K = 1/\theta_K}$$

But without solving, we can just say:

$$\tilde{G}(s) = \mathcal{L}^{-1}\{\varphi(t)\}(s) \implies G(\theta) = \frac{\tilde{G}(1/\theta)}{\theta^2}.$$

All together

$$G(\theta) = \frac{1}{\theta^2}\mathcal{L}^{-1}\left\{e^{-(s_K t)^n}\right\}(1/\theta).$$

1.1 Discrete relaxation times

If we want to solve it we need to make it discrete. So we can write:

$$G(\theta) = \sum_{i=1}^{N} G_i \delta(\theta - \theta_i)$$

so that there are T discrete relaxation times θ_i with weights G_i. Then we have our relaxation function as:

$$\varphi(t) = \sum_{i=1}^{N}\int_0^\infty G_i\delta(\theta - \theta_i)e^{-t/\theta}d\theta = \sum_{i=1}^{N} G_i e^{-t/\theta_i}.$$

Now we can equate this to the Kohlrausch function:

$$e^{-(t/\theta_K)^n} = \sum_{i=1}^{N} G_i e^{-t/\theta_i}.$$

1.2 Finishing the inverse Laplace transform

We have

$$\widetilde{G}(s) = \mathcal{L}^{-1}\left\{e^{-(s_K t)^n}\right\}(s), \qquad s_K = \frac{1}{\theta_K}, \quad 0 < n < 1.$$

Expand the stretched exponential:

$$e^{-(s_K t)^n} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} s_K^{nk} t^{nk}.$$

Use the inverse Laplace transform identity

$$\mathcal{L}^{-1}\{t^a\}(s) = \frac{1}{\Gamma(-a)} s^{-a-1}$$

which can be obtained by analytic continuation of $\mathcal{L}\{t^{\alpha-1}/\Gamma(\alpha)\} = s^{-\alpha}$. With $a = nk$, this gives

$$\mathcal{L}^{-1}\{t^{nk}\}(s) = \frac{1}{\Gamma(-nk)} s^{-nk-1}.$$

Therefore

$$\boxed{\widetilde{G}(s) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} s_K^{nk} \frac{1}{\Gamma(-nk)} s^{-nk-1}.}$$

1.3 Convert back to the relaxation-time spectrum

Recall $\widetilde{G}(s) = G(1/s)/s^2$, hence $G(\theta) = \widetilde{G}(1/\theta)/\theta^2$. Substitute $s = 1/\theta$:

$$\widetilde{G}(1/\theta) = \theta \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \frac{1}{\Gamma(-nk)} (s_K \theta)^{nk}.$$

Thus

$$\boxed{G(\theta) = \frac{1}{\theta} \sum_{k=1}^{\infty} \frac{(-1)^k}{k!} \frac{1}{\Gamma(-nk)} \left(\frac{\theta}{\theta_K}\right)^{nk}, \qquad (0 < n < 1).} \tag{1.3.1}$$

2 2-8 Malkin

Eq. 2.3.11 and its solution show that the Burgers model *describes the behavior of a material with two relaxation times. The same behavior is represented by two parallel*

Maxwell elements with their relaxation times $\theta_1 = n_1/G_1$ and $\theta_2 = n_2/G_2$ where n_1 and G_1 are the viscosity and elastic modulus of the first and n_2 and G_2 of the second Maxwell elements joined in parallel. Calculate the values of the constants of the Burgers model expressed via constants of the two Maxwell elements.

2.1 Burgers Model:

For the total strain rate we have:

$$\dot{\gamma} = \dot{\gamma}_m + \dot{\gamma}_k = \frac{\sigma}{\eta_m} + \frac{\dot{\sigma}}{G_m} + \dot{\gamma}_k.$$

$$\dot{\gamma}_k = \dot{\gamma} - \frac{\sigma}{\eta_m} - \frac{\dot{\sigma}}{G_m}, \quad \ddot{\gamma}_k = \ddot{\gamma} - \frac{\dot{\sigma}}{\eta_m} - \frac{\ddot{\sigma}}{G_m}.$$

The stress across the Maxwell and Kelvin-Voigt elements is the same:

$$\sigma = \sigma_m = \sigma_k \implies \sigma = G_k \gamma_k + \eta_k \dot{\gamma}_k.$$

Now we can differentiate this then substitute Eq. 2.1 into it:

$$\dot{\sigma} = G_k \dot{\gamma}_k + \eta_k \ddot{\gamma}_k.$$

$$\dot{\sigma} = G_k \left(\dot{\gamma} - \frac{\sigma}{\eta_m} - \frac{\dot{\sigma}}{G_m} \right) + \eta_k \left(\ddot{\gamma} - \frac{\dot{\sigma}}{\eta_m} - \frac{\ddot{\sigma}}{G_m} \right).$$

Collecting terms gives:

$$\frac{\eta_k}{G_m} \ddot{\sigma} + \left(1 + \frac{G_k}{G_m} + \frac{\eta_k}{\eta_m} \right) \dot{\sigma} + \frac{G_k}{\eta_m} \sigma = \eta_k \ddot{\gamma} + G_k \dot{\gamma}. \tag{2.1.1}$$

2.2 Two Maxwell Elements in Parallel:

Consider branch 1, a Maxwell element with spring G_1 in series with dashpot η_1. Within this Maxwell element the stress is the same in the spring and dashpot:

$$\sigma_1 = \sigma_{G_1} = \sigma_{\eta_1}, \qquad \sigma_1 = G_1 \gamma_{G_1}, \qquad \sigma_1 = \eta_1 \dot{\gamma}_{\eta_1}.$$

Therefore,

$$\dot{\gamma}_{G_1} = \frac{\dot{\sigma}_1}{G_1}, \qquad \dot{\gamma}_{\eta_1} = \frac{\sigma_1}{\eta_1}.$$

Since the spring and dashpot are in series, strains add in branch 1:

$$\begin{aligned} \dot{\gamma}_1 &= \dot{\gamma}_{G_1} + \dot{\gamma}_{\eta_1} \\ &= \frac{\dot{\sigma}_1}{G_1} + \frac{\sigma_1}{\eta_1}. \end{aligned}$$

Introducing the time-derivative operator $D \equiv \frac{d}{dt}$, this may be written as

$$\dot{\gamma}_1 = \left(\frac{1}{G_1} D + \frac{1}{\eta_1} \right) \sigma_1 = \frac{\eta_1 D + G_1}{G_1 \eta_1} \sigma_1.$$

Similarly for branch 2:

$$\dot{\gamma}_2 = \frac{\dot{\sigma}_2}{G_2} + \frac{\sigma_2}{\eta_2} = \left(\frac{1}{G_2} D + \frac{1}{\eta_2} \right) \sigma_2 = \frac{\eta_2 D + G_2}{G_2 \eta_2} \sigma_2.$$

Since the two branches are in parallel, the total strain rate is the same in both branches:

$$\dot{\gamma} = \frac{\eta_1 D + G_1}{G_1 \eta_1} \sigma_1 = \frac{\eta_2 D + G_2}{G_2 \eta_2} \sigma_2.$$

$$\sigma_1 = \frac{G_1 \eta_1}{\eta_1 D + G_1} \dot{\gamma}, \qquad \sigma_2 = \frac{G_2 \eta_2}{\eta_2 D + G_2} \dot{\gamma}.$$

The total stress is the sum of the stresses in each branch:

$$\begin{aligned}
\sigma &= \sigma_1 + \sigma_2 \\
\sigma &= \frac{G_1 \eta_1}{\eta_1 D + G_1} \dot{\gamma} + \frac{G_2 \eta_2}{\eta_2 D + G_2} \dot{\gamma} \\
\sigma &= \left(\frac{G_1 \eta_1}{\eta_1 D + G_1} + \frac{G_2 \eta_2}{\eta_2 D + G_2} \right) \dot{\gamma} \\
\sigma &= \frac{G_1 \eta_1 (\eta_2 D + G_2) + G_2 \eta_2 (\eta_1 D + G_1)}{(\eta_2 D + G_2)(\eta_1 D + G_1)} \dot{\gamma} \\
(\eta_2 D + G_2)(\eta_1 D + G_1) \sigma &= (G_1 \eta_1 (\eta_2 D + G_2) + G_2 \eta_2 (\eta_1 D + G_1)) \dot{\gamma} \\
(\eta_1 \eta_2 D^2 + (\eta_1 G_2 + \eta_2 G_1) D + G_1 G_2) \, \sigma &= (G_1 \eta_1 \eta_2 D + G_1 \eta_1 G_2 + G_2 \eta_2 \eta_1 D + G_2 \eta_2 G_1) \dot{\gamma} \\
\eta_1 \eta_2 \ddot{\sigma} + (\eta_1 G_2 + \eta_2 G_1) \dot{\sigma} + G_1 G_2 \sigma &= ((G_1 \eta_1 \eta_2 + G_2 \eta_2 \eta_1) D + G_1 \eta_1 G_2 + G_2 \eta_2 G_1) \dot{\gamma}
\end{aligned}$$

$$\eta_1 \eta_2 \ddot{\sigma} + (\eta_1 G_2 + \eta_2 G_1) \dot{\sigma} + G_1 G_2 \sigma = (G_1 \eta_1 \eta_2 + G_2 \eta_2 \eta_1) \ddot{\gamma} + (G_1 \eta_1 G_2 + G_2 \eta_2 G_1) \dot{\gamma}$$

Finally,

$$\frac{\eta_1 \eta_2}{G_1 G_2} \ddot{\sigma} + \frac{(\eta_1 G_2 + \eta_2 G_1)}{G_1 G_2} \dot{\sigma} + \sigma = \frac{\eta_1 (G_1 \eta_2 + G_2 \eta_2)}{G_1 G_2} \ddot{\gamma} + (\eta_1 + \eta_2) \dot{\gamma} \tag{2.2.1}$$

using $\theta_i = \eta_i / G_i$:

$$\theta_1 \theta_2 \ddot{\sigma} + (\theta_1 + \theta_2) \dot{\sigma} + \sigma = (\theta_1 + \theta_2) \ddot{\gamma} + (\theta_1 G_1 + \theta_2 G_2) \dot{\gamma} \tag{2.2.2}$$

gives us a cleaner form.

2.3 Equating Coefficients

From Eq. 2.2.1 we have

$$\frac{\eta_1 \eta_2}{G_1 G_2} \ddot{\sigma} + \frac{(\eta_1 G_2 + \eta_2 G_1)}{G_1 G_2} \dot{\sigma} + \sigma = \frac{\eta_1 \eta_2 (G_1 + G_2)}{G_1 G_2} \ddot{\gamma} + (\eta_1 + \eta_2) \dot{\gamma}$$

From Eq. 2.1.1 we have

$$\frac{\eta_k}{G_m}\,\ddot{\sigma} + \left(1 + \frac{G_k}{G_m} + \frac{\eta_k}{\eta_m}\right)\dot{\sigma} + \frac{G_k}{\eta_m}\,\sigma = \eta_k\,\ddot{\gamma} + G_k\,\dot{\gamma}.$$

So we can compare the two equations, we'll multiply Eq. 2.1.1 by $\frac{\eta_m}{G_k}$:

$$\frac{\eta_k\eta_m}{G_mG_k}\,\ddot{\sigma} + \left(\frac{\eta_m}{G_k} + \frac{\eta_m}{G_m} + \frac{\eta_k}{G_k}\right)\dot{\sigma} + \sigma = \frac{\eta_k\eta_m}{G_k}\,\ddot{\gamma} + \eta_m\,\dot{\gamma}.$$

From left to right, we equate coefficients:

$$\frac{\eta_1\eta_2}{G_1G_2} = \frac{\eta_k\eta_m}{G_mG_k} \tag{2.3.1}$$

$$\frac{(\eta_1G_2 + \eta_2G_1)}{G_1G_2} = \frac{\eta_m}{G_k} + \frac{\eta_m}{G_m} + \frac{\eta_k}{G_k} \tag{2.3.2}$$

$$\frac{\eta_1\eta_2(G_1 + G_2)}{G_1G_2} = \frac{\eta_k\eta_m}{G_k} \tag{2.3.3}$$

$$(\eta_1 + \eta_2) = \eta_m \tag{2.3.4}$$

Firstly we have the trivial result from Eq. 2.3.4:

$$\boxed{\eta_m = \eta_1 + \eta_2}.$$

Theoretically, we also have

$$G_m = G_1 + G_2$$

but we can show this by plugging Eq. 2.3.1 into Eq. 2.3.3:

$$\frac{\eta_1\eta_2}{G_1G_2}(G_1 + G_2) = \frac{\eta_k\eta_m}{G_k}$$
$$\frac{\eta_k\eta_m}{G_mG_k}(G_1 + G_2) = \frac{\eta_k\eta_m}{G_k}$$

$$\boxed{G_m = G_1 + G_2}.$$

Using $\eta_m = \eta_1 + \eta_2$, $G_m = G_1 + G_2$, and

$$\frac{\eta_k}{G_k} = \frac{\eta_1\eta_2(G_1 + G_2)}{G_1G_2(\eta_1 + \eta_2)},$$

we can substitute into Eq. 2.3.2 to solve for G_k:

$$\frac{\eta_m}{G_k} = \frac{\eta_1G_2 + \eta_2G_1}{G_1G_2} - \frac{\eta_1 + \eta_2}{G_1 + G_2} - \frac{\eta_1\eta_2(G_1 + G_2)}{G_1G_2(\eta_1 + \eta_2)}.$$
$$\frac{\eta_m}{G_k} = \frac{(\eta_1G_2 + \eta_2G_1)(G_1 + G_2)(\eta_1 + \eta_2) - G_1G_2(\eta_1 + \eta_2)^2 - \eta_1\eta_2(G_1 + G_2)^2}{G_1G_2(G_1 + G_2)(\eta_1 + \eta_2)}.$$

Now expand the first term in the numerator:

$$\begin{aligned}(\eta_1G_2 + \eta_2G_1)(\eta_1 + \eta_2) &= \eta_1^2G_2 + \eta_1\eta_2G_2 + \eta_1\eta_2G_1 + \eta_2^2G_1 \\ &= \eta_1^2G_2 + \eta_2^2G_1 + \eta_1\eta_2(G_1 + G_2).\end{aligned}$$

So

$$(\eta_1 G_2 + \eta_2 G_1)(G_1 + G_2)(\eta_1 + \eta_2) = (G_1 + G_2)\left[\eta_1^2 G_2 + \eta_2^2 G_1 + \eta_1\eta_2(G_1 + G_2)\right]$$
$$= (G_1 + G_2)(\eta_1^2 G_2 + \eta_2^2 G_1) + \eta_1\eta_2(G_1 + G_2)^2.$$

Substitute this back into the numerator; the $\eta_1\eta_2(G_1 + G_2)^2$ terms cancel:

$$\begin{aligned} \text{numerator} &= \left[(G_1 + G_2)(\eta_1^2 G_2 + \eta_2^2 G_1) + \eta_1\eta_2(G_1 + G_2)^2\right] \\ &\quad - G_1 G_2(\eta_1 + \eta_2)^2 - \eta_1\eta_2(G_1 + G_2)^2 \\ &= (G_1 + G_2)(\eta_1^2 G_2 + \eta_2^2 G_1) - G_1 G_2(\eta_1 + \eta_2)^2. \end{aligned}$$

Expand the remaining pieces:

$$(G_1 + G_2)(\eta_1^2 G_2 + \eta_2^2 G_1) = \eta_1^2 G_1 G_2 + \eta_1^2 G_2^2 + \eta_2^2 G_1^2 + \eta_2^2 G_1 G_2,$$
$$G_1 G_2(\eta_1 + \eta_2)^2 = G_1 G_2(\eta_1^2 + 2\eta_1\eta_2 + \eta_2^2) = \eta_1^2 G_1 G_2 + 2\eta_1\eta_2 G_1 G_2 + \eta_2^2 G_1 G_2.$$

Subtracting gives

$$\text{numerator} = (\eta_1^2 G_2^2 + \eta_2^2 G_1^2) - 2\eta_1\eta_2 G_1 G_2 = (G_1\eta_2 - G_2\eta_1)^2.$$

then

$$\frac{\eta_m}{G_k} = \frac{(G_1\eta_2 - G_2\eta_1)^2}{G_1 G_2(G_1 + G_2)(\eta_1 + \eta_2)}.$$

Since $\eta_m = \eta_1 + \eta_2$, invert to obtain

$$\boxed{G_k = \frac{G_1 G_2\,(G_1 + G_2)\,(\eta_1 + \eta_2)^2}{(G_1\eta_2 - G_2\eta_1)^2}}.$$

Finally, using Eq. 2.3.1 $\eta_k/G_k = \eta_1\eta_2(G_1 + G_2)/[G_1 G_2(\eta_1 + \eta_2)]$, we can solve for η_k:

$$\eta_k = G_k \frac{\eta_k}{G_k} = \left[\frac{G_1 G_2\,(G_1 + G_2)\,(\eta_1 + \eta_2)^2}{(G_1\eta_2 - G_2\eta_1)^2}\right]\left[\frac{\eta_1\eta_2(G_1 + G_2)}{G_1 G_2(\eta_1 + \eta_2)}\right],$$

so

$$\boxed{\eta_k = \frac{\eta_1\eta_2\,(G_1 + G_2)^2\,(\eta_1 + \eta_2)}{(G_1\eta_2 - G_2\eta_1)^2}}.$$

3 3-5 and 3-6 Malkin

3-5: Calculate shear stresses in the flow of liquid through a straight tube if the flow is created by the pressure gradient $\Delta p/L$ (L is the length of a tube).

3-6: Calculate the radial distribution of shear rates and flow velocity of Newtonian liquid (having viscosity η) through a straight tube with radius R.

I decided to combine these because the first first part of 3-6 entirely answers 3-5.

Starting with the *Navier-Stokes equation*:

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u}\right) = -\nabla p + \nabla\cdot\boldsymbol{\sigma}.$$

Evaluating the left-hand side, we have the following due to the steady flow assumption:

$$\frac{\partial \mathbf{u}}{\partial t} = 0.$$

And for the other LHS term:

$$((\mathbf{u}\cdot\nabla)\mathbf{u})_i = u_j\partial_j u_i = u_r\partial_r u_i + u_\theta\tfrac{1}{r}\partial_\theta u_i + u_z\partial_z u_i$$

$u_r = u_\theta = 0$ therefore

$$((\mathbf{u}\cdot\nabla)\mathbf{u})_r = ((\mathbf{u}\cdot\nabla)\mathbf{u})_\theta = 0$$

$\mathbf{u} = u_z(r)$

$$((\mathbf{u}\cdot\nabla)\mathbf{u})_i = u_r\partial_r u_z + u_\theta\tfrac{1}{r}\partial_\theta u_z + u_z\partial_z u_z = 0$$

Therefore the whole LHS of the Navier Stokes is zero. Navier-Stokes equation reduces to

$$\nabla p = \nabla\cdot\boldsymbol{\sigma}.$$

In cylindrical coordinates, the RHS is

$$\nabla\cdot\boldsymbol{\sigma} = \frac{1}{r}\frac{\partial}{\partial r}(r\sigma_{rz}) + \frac{1}{r}\frac{\partial\sigma_{\theta z}}{\partial\theta} + \frac{\partial\sigma_{zz}}{\partial z} = \frac{1}{r}\frac{\partial}{\partial r}(r\sigma_{rz}).$$

All other terms are zero since $\sigma_{\theta z} = 0$ and σ_{zz} is independent of z for steady flow. The pressure gradient is constant, so $\nabla p = \frac{\Delta P}{L}$, and we have

$$\begin{aligned}
\frac{\Delta P}{L} &= \frac{1}{r}\frac{\partial}{\partial r}(r\sigma_{rz}) \\
\frac{\Delta P}{L}r &= \frac{\partial}{\partial r}(r\sigma_{rz}) \\
\int\frac{\Delta P}{L}r\,dr &= \int\frac{\partial}{\partial r}(r\sigma_{rz})\,dr \\
\frac{\Delta P}{L}\frac{r^2}{2} + C_1 &= r\sigma_{rz}.
\end{aligned}$$

Solving for the shear stress gives

$$\sigma_{rz} = \frac{\Delta P}{2L}r + \frac{C_1}{r}.$$

The stress must remain finite at the centerline $r = 0$, which requires $C_1 = 0$. Therefore

$$\boxed{\sigma_{rz} = \frac{\Delta P}{2L}r}.$$

This is a key result: the shear stress σ_{rz} is proportional to the radius r and does not depend on the viscosity η.

Now we can use the constitutive relation for a Newtonian fluid $\sigma_{rz} = \eta\dot{\gamma}_{rz}$ to find the shear rate distribution:

$$\dot{\gamma}_{rz} = \frac{\sigma_{rz}}{\eta} = \frac{\Delta P}{2\eta L}r.$$

Integrating the shear rate gives the velocity distribution:

$$\begin{aligned} u_z(r) &= \int \dot{\gamma}_{rz}\,dr \\ &= \int \frac{\Delta P}{2\eta L}r\,dr \\ &= \frac{\Delta P}{4\eta L}r^2 + C_2. \end{aligned}$$

Applying the no-slip boundary condition $u_z(R) = 0$ gives $C_2 = -\frac{\Delta P}{4\eta L}R^2$, so the velocity profile becomes

$$\boxed{u_z(r) = \frac{\Delta P}{4\eta L}\left(r^2 - R^2\right)}.$$

Additional question 1. *Calculate the volume output, Q, for the flow of Newtonian liquid.*

The flow rate is given by

$$Q = \int \mathbf{u} \cdot d\mathbf{S}.$$

All of the flow field $\mathbf{u}$ has the same direction as the outlet S so

$$Q = \int u\,dS.$$

The surface is just the cross sectional area of the pipe:

$$Q = \int_0^R \int_0^{2\pi} u\,r\,dr\,d\theta = 2\pi \int_0^R u\,r\,dr$$

$$Q = 2\pi\frac{\Delta P}{4\eta L}\int_0^R (r^2 - R^2)rdr$$

$$\int_0^R r^3dr = \frac{R^4}{4}, \quad R^2\int_0^R rdr = \frac{R^4}{2}$$

$$Q = 2\pi\frac{\Delta P}{4\eta L}\left(\frac{R^4}{4} - \frac{R^4}{2}\right) = \boxed{\frac{\pi\Delta P R^4}{8\eta L}}$$

Additional question 2. *Express maximum shear rate via volume output.* From earlier we have:

$$\sigma_{rx} = \frac{\Delta P r}{2L}.$$

$r \in [0, R]$, therefore:

$$\sigma_{\max} = \frac{\Delta P R}{2L}.$$

$$Q = \frac{\pi R^3}{4\eta}\frac{\Delta P R}{2L} = \frac{\pi R^3}{4\eta}\sigma_{\max} \implies \sigma_{\max} = \boxed{\frac{4\eta Q}{\pi R^3}}$$

Additional question 3. *Is the last expression valid for a liquid with arbitrary rheological properties?*

No, in our derivation we assumed that $\sigma \propto \dot{\gamma}$. If we violate that assumption, we would need to modify our equation accordingly.

4 3-9 Malkin

Analyze the flow of a viscoplastic ("Bingham-type") liquid through a straight tube of radius R. Find radial stress and velocity distributions and calculate volume output as a function of the pressure gradient.

According to Malkin, the *Bingham Equation* is

$$\sigma = \sigma_Y + \mu\dot{\gamma}, \tag{4.0.1}$$

where σ_Y is the yield stress and μ is the plastic viscosity. Or equivalently,

$$\dot{\gamma} = \begin{cases} 0 & \text{if } \sigma < \sigma_Y \\ \frac{\sigma - \sigma_Y}{\mu} & \text{if } \sigma \geq \sigma_Y \end{cases}.$$

Next we want we want to find the shear stress distribution σ_{ij} as a function of r. Starting with the *Navier-Stokes equation*:

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u}\right) = -\nabla p + \nabla \cdot \boldsymbol{\sigma}.$$

Evaluating the left-hand side, we have the following due to the steady flow assumption:

$$\frac{\partial \mathbf{u}}{\partial t} = 0.$$

And for the other LHS term:

$$((\mathbf{u} \cdot \nabla)\mathbf{u})_i = u_j \partial_j u_i = u_r \partial_r u_i + u_\theta \tfrac{1}{r}\partial_\theta u_i + u_z \partial_z u_i$$

$u_r = u_\theta = 0$ therefore

$$((\mathbf{u} \cdot \nabla)\mathbf{u})_r = ((\mathbf{u} \cdot \nabla)\mathbf{u})_\theta = 0$$

$\mathbf{u} = u_z(r)$

$$((\mathbf{u} \cdot \nabla)\mathbf{u})_i = u_r \partial_r u_z + u_\theta \tfrac{1}{r}\partial_\theta u_z + u_z \partial_z u_z = 0$$

Therefore the whole LHS of the Navier Stokes is zero. Navier-Stokes equation reduces to

$$\nabla p = \nabla \cdot \boldsymbol{\sigma}.$$

In cylindrical coordinates, the RHS is

$$\nabla \cdot \boldsymbol{\sigma} = \frac{1}{r}\frac{\partial}{\partial r}(r\sigma_{rz}) + \frac{1}{r}\frac{\partial \sigma\theta z}{\partial \theta} + \frac{\partial \sigma_{zz}}{\partial z} = \frac{1}{r}\frac{\partial}{\partial r}(r\sigma_{rz}).$$

All other terms are zero since $\sigma_{\theta z} = 0$ and σ_{zz} is independent of z for steady flow. The pressure gradient is constant, so $\nabla p = \frac{\Delta P}{L}$, and we have

$$\begin{aligned}
\frac{\Delta P}{L} &= \frac{1}{r}\frac{\partial}{\partial r}(r\sigma_{rz}) \\
\int \frac{\Delta P}{L} r\, dr &= \int \frac{\partial}{\partial r}(r\sigma_{rz})\, dr \\
\frac{\Delta P}{L}\frac{r^2}{2} + C_1 &= r\sigma_{rz} \\
\sigma_{rz} &= \frac{\Delta P}{2L} r + \frac{C_1}{r}.
\end{aligned}$$

The stress must remain finite at the centerline $r = 0$, which requires $C_1 = 0$. Therefore

$$\boxed{\sigma_{rz} = \frac{\Delta P}{2L} r}.$$

Flow only occurs when $\sigma(r) \geq \sigma_Y$, so there is a region in the center of the tube where $r < r_0$ for which there is no shearing. The radius r_0 is given by:

$$\sigma(r_0) = \sigma_Y \implies r_0 = \frac{2L}{\Delta P}\sigma_Y.$$

Recalling that for our "Bingham fluid" $\sigma = \sigma_Y + \mu\dot{\gamma}$, we can write $\dot{\gamma}$ as

$$\dot{\gamma} = \frac{\sigma - \sigma_Y}{\mu} = \frac{\Delta P}{2\mu L} r - \frac{\sigma_Y}{\mu}.$$

Adding the requirement that $\dot{\gamma} = 0$ for $r < r_0$ gives the full shear rate distribution:

$$\boxed{\dot{\gamma} = \begin{cases} 0 & r < r_0 \\ \frac{\Delta P}{2\mu L} r - \frac{\sigma_Y}{\mu} & r \geq r_0 \end{cases}}.$$

As you increase r, the velocity u_z decreases, so the shear rate $\dot{\gamma}$ is negative. Therefore,

$$\dot{\gamma} = -\frac{du_z}{dr} \implies du_z = -\dot{\gamma} dr.$$

Now we can integrate to get the velocity distribution in the region $r \geq r_0$:

$$u_z = -\int \dot{\gamma} dr = -\int \left(\frac{\Delta P}{2\mu L} r - \frac{\sigma_Y}{\mu} \right) dr = -\frac{\Delta P}{4\mu L} r^2 + \frac{\sigma_Y}{\mu} r + C.$$

From the zero slip condition at the wall, we have

$$0 = -\frac{\Delta P}{4\mu L} R^2 + \frac{\sigma_Y}{\mu} R + C \implies C = -\frac{\sigma_Y}{\mu} R + \frac{\Delta P}{4\mu L} R^2.$$

Therefore, the velocity distribution in the region $r \geq r_0$ is

$$u_z(r) = \frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r).$$

For the region $r < r_0$, there is no shearing, so the velocity is constant. The value of this constant is given by the velocity at $r = r_0$:

$$u_z(r) = \frac{\Delta P}{4\mu L}(R^2 - r_0^2) - \frac{\sigma_Y}{\mu}(R - r_0).$$

Recall that

$$r_0 = \frac{2L}{\Delta P}\sigma_Y \implies \sigma_Y = \frac{\Delta P}{2L} r_0,$$

so

$$\frac{\sigma_Y}{\mu}(R - r_0) = \frac{\Delta P}{2\mu L} r_0 (R - r_0).$$

Substituting this into the velocity expression gives

$$\begin{aligned} u_z(r) = \frac{\Delta P}{4\mu L}(R^2 - r_0^2) - \frac{\Delta P}{2\mu L} r_0 (R - r_0) &= \frac{\Delta P}{4\mu L}\left[(R^2 - r_0^2) - 2r_0(R - r_0)\right]. \\ &= \frac{\Delta P}{4\mu L}(R^2 - 2Rr_0 + r_0^2) = \frac{\Delta P}{4\mu L}(R - r_0)^2. \end{aligned}$$

Then the full velocity distribution is

$$\boxed{u_z(r) = \begin{cases} \frac{\Delta P}{4\mu L}(R - r_0)^2 & r < r_0 \\ \frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r) & r \geq r_0 \end{cases}}.$$

Finally, we can calculate the volume flow Q by integrating the velocity profile across the cross-sectional area of the tube. Since $d\mathbf{A} = r\, dr\, d\theta\, \hat{z}$ and $\mathbf{u} = u_z(r)\hat{z}$, we have

$$\begin{aligned} Q &= 2\pi \int_0^R u_z(r)\, r\, dr \\ &= 2\pi \left[\int_0^{r_0} \frac{\Delta P}{4\mu L}(R - r_0)^2 r\, dr + \int_{r_0}^R \left(\frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r) \right) r\, dr \right]. \end{aligned}$$

For the region $(0 \le r \le r_0)$:

$$\begin{aligned}
\int_0^{r_0} \frac{\Delta P}{4\mu L}(R - r_0)^2 r\, dr &= \frac{\Delta P}{4\mu L}(R - r_0)^2 \int_0^{r_0} r\, dr \\
&= \frac{\Delta P}{4\mu L}(R - r_0)^2 \left[\frac{r^2}{2}\right]_0^{r_0} \\
&= \frac{\Delta P}{8\mu L}(R - r_0)^2 r_0^2.
\end{aligned}$$

For the region $(r_0 \le r \le R)$:

$$\int_{r_0}^{R} \left(\frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r)\right) r\, dr = \int_{r_0}^{R} \left[\frac{\Delta P}{4\mu L}(R^2 r - r^3) - \frac{\sigma_Y}{\mu}(Rr - r^2)\right] dr.$$

Integrate term-by-term:

$$\begin{aligned}
\int R^2 r\, dr &= R^2 \frac{r^2}{2}, \qquad & \int r^3\, dr &= \frac{r^4}{4}, \\
\int Rr\, dr &= R\frac{r^2}{2}, \qquad & \int r^2\, dr &= \frac{r^3}{3}.
\end{aligned}$$

Plugging the integrals back in gives

$$\begin{aligned}
&\int_{r_0}^{R} \left(\frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r)\right) r\, dr \\
&= \left[\frac{\Delta P}{4\mu L}\left(R^2\frac{r^2}{2} - \frac{r^4}{4}\right) - \frac{\sigma_Y}{\mu}\left(R\frac{r^2}{2} - \frac{r^3}{3}\right)\right]_{r_0}^{R}.
\end{aligned}$$

Evaluate at $r = R$:

$$\begin{aligned}
&\frac{\Delta P}{4\mu L}\left(\frac{R^4}{2} - \frac{R^4}{4}\right) - \frac{\sigma_Y}{\mu}\left(\frac{R^3}{2} - \frac{R^3}{3}\right) \\
&= \frac{\Delta P}{16\mu L}R^4 - \frac{\sigma_Y}{6\mu}R^3.
\end{aligned}$$

Evaluate at $r = r_0$:

$$\frac{\Delta P}{4\mu L}\left(R^2\frac{r_0^2}{2} - \frac{r_0^4}{4}\right) - \frac{\sigma_Y}{\mu}\left(R\frac{r_0^2}{2} - \frac{r_0^3}{3}\right).$$

Combining both regions gives the final expression for Q:

$$Q = 2\pi\left[\frac{\Delta P}{8\mu L}(R - r_0)^2 r_0^2 + \frac{\Delta P}{16\mu L}R^4 - \frac{\sigma_Y}{6\mu}R^3 - \frac{\Delta P}{4\mu L}\left(R^2\frac{r_0^2}{2} - \frac{r_0^4}{4}\right) + \frac{\sigma_Y}{\mu}\left(R\frac{r_0^2}{2} - \frac{r_0^3}{3}\right)\right]$$

where $r_0 = \frac{2L\sigma_Y}{\Delta P}$, in terms of the pressure gradient $\Delta P/L$, the yield stress σ_Y, the plastic viscosity μ, and the tube radius R.

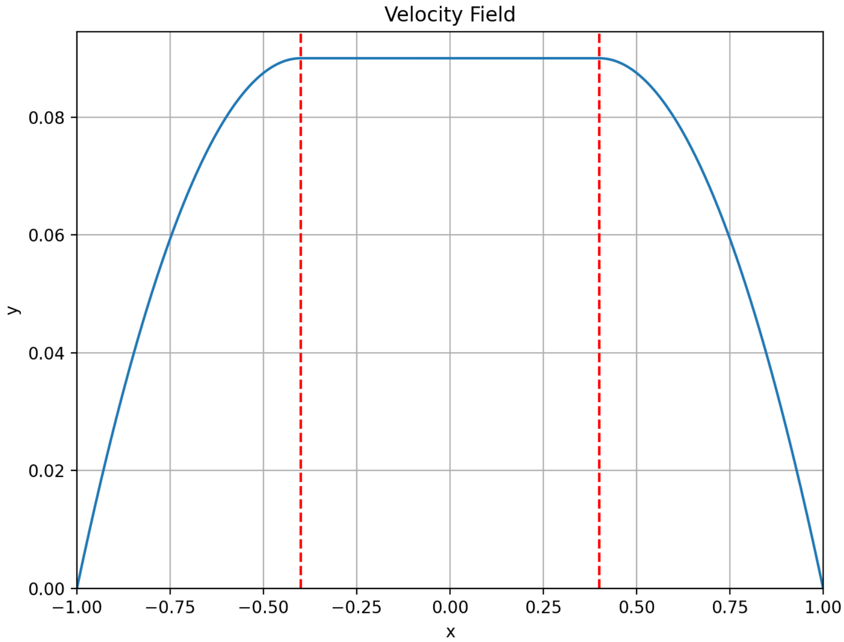

Figure 1: Velocity field for a Bingham fluid flowing through a tube.

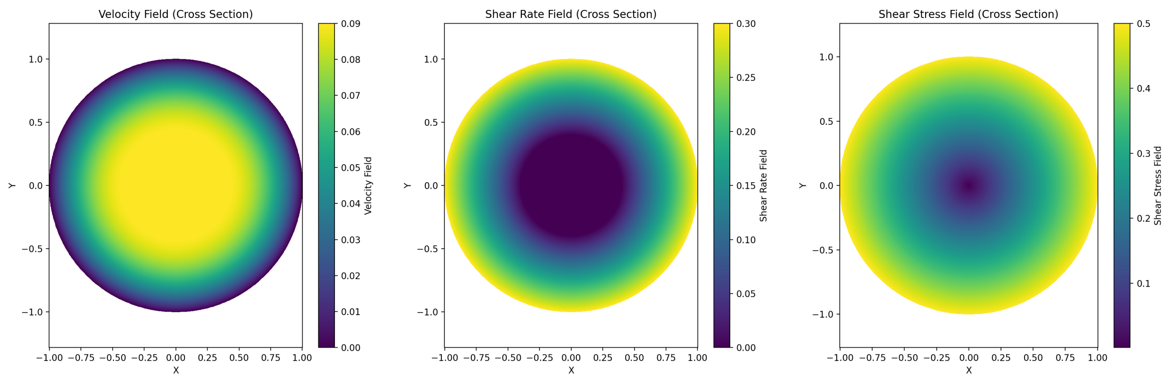

Figure 2: Cross-sectional profiles of velocity, shear rate, and shear stress for a Bingham fluid flowing through a tube. The region near the center of the tube (where $r < r_0$) has zero shear rate. The velocity profile is flat in this region and then decreases towards the walls of the tube. The shear stress increases linearly with radius and exceeds the yield stress σ_Y at $r = r_0$.

5 5-7 Malkin

In the principal scheme of measuring viscoelastic properties of the material, a spring is used (Fig. 5.7.1) that is not ideal elastic but viscoelastic (has some losses in deformation).

How do you calculate the viscoelastic properties of the material under investigation?

The solution for the driving oscillator x_1 and the top plate x_2 are:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 e^{i\varphi} \end{bmatrix} e^{i\omega t}$$

For the top plate we have:

$$m\ddot{x}_2 + Zx_2 + \frac{S}{h}G^*(x_2 + x_1) = 0$$

where m is the mass of the top plate, Z is the modulus of the spring, G_* is the modulus of the material, S is the surface of the plate and h is the height of the plate. If the spring has no loss modulus, Z is purely real. In this case we consider a spring with a non-zero loss modulus, therefore $Z \to Z^* = Z' + iZ''$ had to be a complex number.

$$m\ddot{x}_2 + Z^*x_2 + \frac{S}{h}G^*(x_2 + x_1) = 0$$

Substituting $x_1 = c_1 e^{i\omega t}$, $x_2 = c_2 e^{i\varphi} e^{i\omega t}$ and $\ddot{x}_2 = -c_2\omega^2 e^{i\varphi} e^{i\omega t}$, then solving for G^*:

$$-mc_2\omega^2 e^{i\varphi} + c_2 Z^* e^{i\varphi} + \frac{S}{h}G^*(c_2 e^{i\varphi} + c_1) = 0$$

$$\frac{S}{h}G^*(c_2 e^{i\varphi} + c_1) = mc_2\omega^2 e^{i\varphi} - c_2 Z^* e^{i\varphi}$$

$$G^* = \frac{h}{S} \cdot \frac{mc_2\omega^2 e^{i\varphi} - c_2 Z^* e^{i\varphi}}{c_2 e^{i\varphi} + c_1}$$

$$G^* = \frac{h}{S} \cdot \frac{m\omega^2 - Z^*}{1 + \frac{c_1}{c_2} e^{-i\varphi}}$$

For simplicity, let $r = c_1/c_2$ Separating real and imaginary components:

$$G^* = \frac{h}{S} \cdot \frac{m\omega^2 - Z' - iZ''}{(1 + r\cos\varphi) - ir\sin\varphi}$$

$$G^* = \frac{h}{S} \cdot \frac{[(m\omega^2 - Z')(1 + r\cos\varphi) - Z''r\sin\varphi] + i\,[Z''(1 + r\cos\varphi) - (m\omega^2 - Z')r\sin\varphi]}{(1 + r\cos\varphi)^2 + r^2\sin^2\varphi}$$

$$G' = \frac{h}{S} \cdot \frac{(m\omega^2 - Z')(1 + r\cos\varphi) - Z''r\sin\varphi}{(1 + r\cos\varphi)^2 + r^2\sin^2\varphi}$$

$$G'' = \frac{h}{S} \cdot \frac{Z''(1 + r\cos\varphi) - (m\omega^2 - Z')r\sin\varphi}{(1 + r\cos\varphi)^2 + r^2\sin^2\varphi}$$

The loss tangent $\delta = G''/G'$ is

$$\delta = \frac{Z''(1 + r\cos\varphi) - (m\omega^2 - Z')r\sin\varphi}{(m\omega^2 - Z')(1 + r\cos\varphi) - Z''r\sin\varphi}$$